

Stereo Image Coding for Machines With Joint Visual Feature Compression

Dengchao Jin , Jianjun Lei , Senior Member, IEEE, Bo Peng , Member, IEEE, Zhaoqing Pan , Senior Member, IEEE, Bo Zhao , Nam Ling , Life Fellow, IEEE, and Qingming Huang , Fellow, IEEE

Abstract—2D image coding for machines (ICMs) has achieved great success in coding efficiency, while less effort has been devoted to stereo image fields. To promote the efficiency of stereo image compression (SIC) and intelligent analysis, the stereo image coding for machines (SICMs) is formulated and explored in this article. More specifically, a machine vision-oriented stereo feature compression network (MVSFC-Net) is proposed for SICM, where the stereo visual features are effectively extracted, compressed, and transmitted for 3D visual task. To efficiently compress stereo visual features in MVSFC-Net, a stereo multi-scale feature compression (SMFC) module is designed to gradually transform sparse stereo multi-scale features into compact joint visual representations by removing spatial, inter-view, and cross-scale redundancies simultaneously. Experimental results show that the proposed MVSFC-Net obtains superior compression efficiency as well as 3D visual task performance, when compared with the existing ICM anchors recommended by MPEG and the state-of-the-art SIC method.

Impact Statement—With the rapid development of stereo image technologies and artificial intelligence, stereo images have been widely used in many fields, such as autonomous driving, smart city and intelligent industry. However, due to the inherent limitations of storage space and transmission bandwidth, the stereo images need to be efficiently compressed to better balance 3D visual analysis performance and transmission cost. In this article, the SICM is explored. Specifically, a novel MVSFC-Net is proposed to achieve high-efficiency SICM. In particular, a SMFC module is designed to gradually transform sparse stereo multi-scale features into compact joint visual representations by removing spatial, inter-view, and cross-scale redundancies simultaneously. It is hoped the proposed MVSFC-Net would benefit the development of the learned SIC field and be helpful to promote 3D visual analysis services under the limited capabilities of stereo image storage and transmission.

Index Terms—Deep neural network, image compression for machines, intelligent coding, machine vision-oriented stereo feature compression network (MVSFC-Net), stereo image, stereo image coding for machines (SICMs).

I. INTRODUCTION

WITH the rapid development of stereo image technologies and artificial intelligence, enormous intelligence applications (including Smart City, Autonomous Driving, Intelligent Industry, Cross-domain Stereo Vision, and so on) have sprouted to replace traditional man-powered pipelines [1], [2], [3]. To complete various 3D visual tasks, these intelligence applications generally need to efficiently handle and analyze massive stereo images, resulting in a huge burden on the data transmission and storage system [4], [5], [6]. For instance, to achieve accurate obstacle avoidance in autonomous driving, stereo images captured by on-board binocular cameras are commonly transmitted to cloud platforms for enhancing 3D visual analysis accuracy [7]. In this case, it is crucial to boost stereo image transmission efficiency without compromising visual analysis performance, thereby promoting the real-time precise decision-making capabilities of autonomous driving systems. Consequently, it is highly desirable to develop high-efficiency stereo image coding for machines (SICMs).

Aiming to reconstruct high-quality stereo images under bitrate constraints, traditional stereo image coding (SIC) methods [8] have been developed over several decades, and the coding efficiency has been improved dramatically. These methods generally employed a traditional hybrid coding framework, where the signal redundancies are removed by multiple hand-crafted technologies, such as intra-prediction, inter-view prediction, and entropy coding [9], [10]. To achieve end-to-end rate-distortion (RD) optimization, recent methods applied deep learning technologies into SIC to further boost compression efficiency [11], [12]. Nevertheless, the aforementioned methods generally focused on reconstructing stereo images for human visual systems (HVS). Since pixel-level signal fidelity rather than high-level semantic fidelity is emphasized, these SIC methods are shown suboptimal 3D visual task performance in SICM, especially at low bitrates.

In recent years, 2D image coding for machines (ICMs) [13], [14] has attracted more and more attention. From the perspective of compressing the source image or visual features, existing ICM methods can be categorized into two pipelines,

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Dengchao Jin, Jianjun Lei, Bo Peng, Zhaoqing Pan, and Bo Zhao are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: jdc3159761141@tju.edu.cn; jjlei@tju.edu.cn; bpeng@tju.edu.cn; zqpan3-c@my.cityu.edu.hk; bzzb@tju.edu.cn).

Nam Ling is with the Department of Computer Science and Engineering, Santa Clara University, Santa Clara, CA 95053 USA (e-mail: nling@scu.edu).

Qingming Huang is with the School of Computer Science and Technology, University of Chinese Academy of Sciences, Beijing 101408, China (e-mail: qmhuang@ucas.ac.cn).

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i.e., compress-then-analyze (CTA) [15], [16], and analyze-then-compress (ATC) [17], [18]. As for the CTA pipeline, the source image is compressed and reconstructed for visual analysis. To facilitate both compression efficiency and visual analysis performance, researchers have concentrated on reconstructing images that are more suitable for subsequent visual tasks, such as latent space masking [16], joint training with visual loss [19], [20], [21], and so on. Different from the CTA pipeline compressing the source image, the ATC pipeline focuses on compressing visual features which are expected to have lower information entropy than the whole high-quality images. In specific, representative visual features [17], [18] are extracted at the front-end, and then compressed as well as transmitted to the service-end for visual analysis. As such, the ATC pipeline can effectively reduce the sizes of bit-stream and decoding computational complexity. However, existing methods mainly focus on ICM, unprecedented challenges remain in the exploration of SICM.

In contrast to 2D images, stereo images enable a dynamic scene visualization with more viewpoints, which leads to a significant increase in the data volume [22], [23], [24]. Apart from removing the spatial redundancies as the ICM, the inter-view redundancies between different viewpoints should also be considered in the SICM to further improve compression efficiency and 3D visual analysis performance. As a result, applying the aforementioned ICM approaches directly to SICM struggles to achieve superior performance, due to no explicit exploration of the inter-view correlations between viewpoints.

To this end, this article proposed a machine vision-oriented stereo feature compression network (MVSFC-Net) for SICM, which extends the previous ATC approaches to a stereo version with a specific cross-scale redundancy removal (CRR) mechanism. To ensure the performance of both compression and visual analysis, the stereo visual features are efficiently extracted, compressed, and transmitted in the MVSFC-Net. Unlike existing ICM and SIC methods that can only reduce partial redundant information in stereo visual features, the key in the MVSFC-Net lies in the joint removal of spatial, cross-scale, and inter-view redundancies among these sparse stereo features. Thus, a stereo multi-scale feature compression (SMFC) module is proposed, where the compact joint visual representation of the stereo features is progressively generated for higher compression efficiency. The contributions are summarized as follows.

- 1) To promote stereo image compression efficiency and 3D visual analysis performance, a MVSFC-Net is proposed. As far as we know, the proposed MVSFC-Net is the first exploration of the SICM.
- 2) A SMFC module is proposed for the stereo visual features compression, in which the intra, inter-view, as well as cross-scale redundancies are simultaneously removed to obtain compact joint visual representation for high-efficiency compression.
- 3) Experimental results prove that the proposed MVSFC-Net outperforms the existing ICM anchors recommended by MPEG and the state-of-the-art SIC method by a large margin in terms of the compression efficiency for 3D visual task performance.

The remainder of the article is organized as follows. Section II systematically reviews the related works on SIC and ICM. After that, Section III formulates the SICM and introduces the proposed MVSFC-Net for SICM in detail. In Section IV, experimental results and analysis are reported. Finally, Section V concludes this article.

II. RELATED WORK

A. SIC for HVS

During the past decades, with the aim to reconstruct high-fidelity stereo images for HVS with the restriction of bitrate, SIC has been widely studied for efficient stereo image storage and transmission [25], [26], [27]. Different from 2D image compression, the SIC generally focuses on removing inter-view redundancies by exploiting the correlation between stereo images. For instance, on the basis of high efficiency video coding (HEVC), the multi-view HEVC (MV-HEVC) [28] significantly improves the SIC performance by extending multi-view coding technologies, such as disparity-compensated prediction (DCP) [29]. To improve the DCP accuracy, Wong et al. [27] considered the horizontal scaling and shearing (HSS) deformations in stereo images and introduced HSS-based DCP to improve SIC efficiency.

With the growing success of deep learning, several researchers have exploited neural networks to construct the learned SIC framework. Compared to traditional methods, learned SIC methods can boost the compression efficiency by jointly optimizing the whole framework with the RD cost. For instance, Lei et al. [11] employed bi-directional coding mechanism in SIC and achieved impressive coding performance improvement. Wodlinger et al. [12] proposed SIC with latent shifts and stereo attention (SASIC), where the encoded left-image latent representation is shifted to the right-image as an inter-view reference prior. Deng et al. [30] designed a mask SIC network termed MASIC, which applies the homography matrix to align the left image with the right image, thereby only compressing the residual information to save bitrates. Liu et al. [31] proposed a bi-directional stereo compression (BiSIC) method with a cross-dimensional entropy model, where multiple conditioning priors are integrated to improve the efficiency of entropy coding. Wodlinger et al. [32] presented epipolar cross attention for SIC (ECSIC), which jointly compresses the left and right images via the stereo cross attention mechanism.

Despite significant progress on SIC for human perceptual, it is undesirable to directly apply the above methods for SICM, due to the emphasized pixel-level signal fidelity rather than high-level semantic fidelity for visual analysis.

B. ICMs

In the era of deep learning, to promote image compression efficiency for visual tasks, many efforts have been devoted to ICM based on CTA pipeline and ATC pipeline.

As for the CTA pipeline, the source image is compressed at the front-end and reconstructed at the decoder side for visual analysis. To reduce bitrate while maintaining visual task performance, several researchers focused on masking out information

that is not required for the visual task. For instance, Fischer et al. [16] proposed a latent space masking network (LSM-Net) to discriminatively squeeze out the non-salient components in the compact latent representation. Another promising solution is to enhance the image coding framework with the visual task-related loss optimization. Concretely, Fischer et al. [33] replaced conventional pixel-level rate-distortion optimization (RDO) with a standard-compliant feature-based RDO (FRDO) for VVC to improve the object detection accuracy. Gao et al. [19] proposed semantics-oriented metrics to encourage the reconstructed images more suitable for various visual tasks. Patwa et al. [20] incorporated a cross-entropy loss term in the end-to-end image compression framework to improve the image classification performance. Wang et al. [21] proposed to optimize the image compression network with the combination of visual task loss and the RD loss. Although these methods obviously improve the visual task performance under bitrate constraints for CTA pipeline, redundancy information for machines in the reconstructed color images is inevitably maintained, which prevents further improvements for coding performance.

In the ATC pipeline, the compact features are first extracted at the front-end, and then encoded as well as transmitted to the service end for visual analysis. In particular, MPEG has designed a series of visual feature descriptors [34], [35] in CDVS [36] for visual search and video analysis. Choi and Bajić [17] developed a collaborative intelligence for mobile applications, where the deep features are near-lossless and lossy compressed for object detection. To enhance the feature generalization for multiple visual tasks, Feng et al. [18] proposed to obtain omnipotent features by taking advantage of self-supervised learning technologies. Chen et al. [37] transferred a Transformer-based image coding network from human vision to multiple machine vision tasks and achieved impressive compression efficiency. Shen et al. [38] proposed novel parallel task-prompts for ICM, which are adaptable to diverse machine vision tasks without the necessity of developing new networks or training from scratch. Liu et al. [39] proposed an efficient feature compression network, which explores the correlation between features of different scales, and utilizes the encoded small-scale features to predict large-scale features. Sun et al. [40] proposed to separately represent the object in a semantically structured bitstream, where visual tasks are performed on the partial bitstream to reduce the decoding computational complexity and the transferred data. To fulfill the needs for machine vision and human vision jointly, several efforts have been devoted to developing a scalable coding framework [41], [42], [43], where an additional bitstream is designed to ensure signal fidelity for HVS. For instance, Yang et al. [41] proposed a face image coding framework, where the compact structure representations are extracted to be the basic layer for machine vision, and the color representation is served as an enhanced layer to generate images for human vision. Wang et al. [42] proposed a collaborative visual information representation to investigate interactions between the visual feature for machine vision and the texture for human vision. Huang et al. [43] proposed an efficient human-machine friendly video coding scheme (HMFVC), where the semantic information for

machine vision is efficiently exploited to enhance the video reconstruction quality for human vision.

However, as far as we know, no previous research on SICM has been reported. Since the inter-view correlations between two viewpoints have not been explicitly explored, it is not satisfactory to apply the aforementioned ICM methods for SICM.

III. THE PROPOSED METHOD

In this section, we elaborate the proposed method for SICM in detail. First, the formulations of SICM and the overall framework of MVSFC-Net are systematically introduced. Second, the proposed SMFC module is illustrated. Finally, the loss functions and implementation details are described.

A. Formulations of SICM and Framework of MVSFC-Net

To promote the SIC efficiency toward 3D machine vision rather than human vision, the SICM is formulated and researched in this article. Formally, the SICM aims to maximize the performance of machine tasks under the constraints on the bitrate, the formulations can be summarized as follows:

$$\arg \max_{\theta} P_{\text{task}}, \text{ s.t. } \sum_{i=1}^n R_i \leq R_{\text{max}} \quad (1)$$

where P_{task} denotes visual task performance, R_i is the bitrate consumption of a viewpoint, n represents the number of viewpoints, R_{max} denotes the bitrate limit, and θ represents the parameters of the network architecture.

To solve the above SICM problem, the MVSFC-Net is correspondingly designed in this article. In the MVSFC-Net, 3D object detection is employed as the 3D visual task, as it is an important research topic in computer vision [44], [45], [46]. Fig. 1 illustrates the framework of MVSFC-Net. As shown in the figure, the MVSFC-Net is composed of stereo feature extraction module, SMFC module, and visual analysis module. Specifically, the stereo feature extraction module is utilized to extract representative visual features, which are capable enough to represent the 3D scene characteristics for the 3D object detection task, while having less information entropy than the full-resolution stereo image. Generally, the high-resolution feature maps from earlier layers are supposed to preserve more original details (i.e., texture and contours of the object), while low-resolution feature maps from deeper layers can capture semantic and contexture information. Accordingly, to ensure satisfactory machine task performance, the stereo multi-scale feature $\{f_L^i, f_R^i | i \in 0, 1, 2\}$ from different layers in the stereo feature extraction module are served as encoding information in the MVSFC-Net. After the stereo feature extraction module, the SMFC module is designed to efficiently compress $\{f_L^i, f_R^i | i \in 0, 1, 2\}$ by exploiting complex correspondences among these features. Finally, the visual analysis module is employed to produce the 3D object detection result based on the reconstructed stereo multi-scale features $\{\hat{f}_L^i, \hat{f}_R^i | i \in 0, 1, 2\}$. Owing to compressing and transmitting representative visual features rather than pixel-level source stereo image, the proposed MVSFC-Net could obtain higher detection accuracy with lower bitrate consumption and computational complexity.

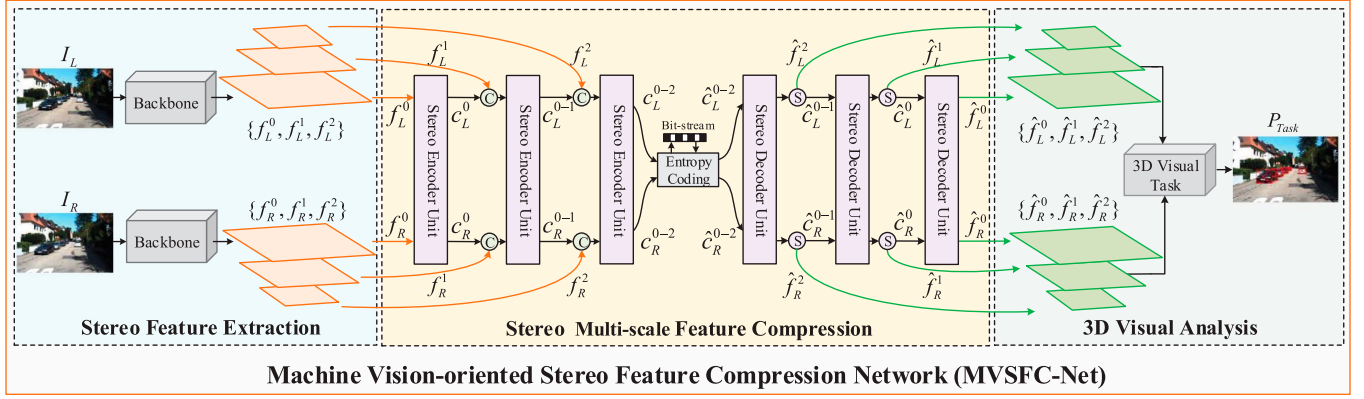


Fig. 1. Architecture of the proposed MVSFC-Net, where $\oplus$ denotes channel-wise concatenation and $\otimes$ represents channel-wise split. For the encoding of stereo images $\{I_L, I_R\}$, the stereo feature extraction module is first applied to obtain the stereo multi-scale features $\{f_L^i, f_R^i | i \in 0, 1, 2\}$. Then, the $\{f_L^i, f_R^i | i \in 0, 1, 2\}$ are efficiently compressed by the proposed SMFC module. Finally, the visual analysis module deployed at service-end is utilized to perform vision task based on reconstructed stereo multi-scale features $\{\hat{f}_L^i, \hat{f}_R^i | i \in 0, 1, 2\}$.

B. SMFC Module

Stereo multi-scale features have higher dimensional data sizes but less information entropy, thus effectively removing redundancies among these sparse features is crucial for MVSFC-Net. Therefore, the SMFC module is proposed to efficiently compress the stereo multi-scale features, where the spatial, inter-view, and cross-scale redundancies are removed simultaneously.

The architecture of the SMFC module is illustrated in Fig. 1. As shown in the figure, taking the sparse stereo multi-scale features $\{f_L^i, f_R^i | i \in 0, 1, 2\}$ as input, the SMFC module first converges the representative information among multi-scale features and generates the compact joint visual representation $\{c_L^{0-2}, c_R^{0-2}\}$. After that, the $\{c_L^{0-2}, c_R^{0-2}\}$ are entropy encoded into bit-stream, which is then decoded in the decoder side to progressively recover the reconstructed stereo multi-scale features $\{\hat{f}_L^i, \hat{f}_R^i | i \in 0, 1, 2\}$ for further visual analysis.

At the beginning of the SMFC module, the first low-level stereo feature $\{f_L^0, f_R^0\}$ are fed into a stereo encoder unit to obtain a low-level compact visual representation $\{c_L^0, c_R^0\}$, i.e.,

$$\{c_L^0, c_R^0\} = \text{SEU}(f_L^0, f_R^0) \quad (2)$$

where $\text{SEU}(\cdot)$ denotes the stereo encoder unit, which is composed of two downsampling convolutions and a bi-directional contextual transform [11]. In particular, the downsampling convolution is applied to reduce the spatial resolution for removing spatial redundancies, while the bi-directional contextual transform is utilized to remove inter-view redundancies by interacting mutual information between $\{f_L^0, f_R^0\}$. Since the outputs of different layers still describe the same dynamic scenes, the medium-level stereo features $\{f_L^1, f_R^1\}$ are jointly compressed with the low-level compact representation $\{c_L^0, c_R^0\}$ to remove cross-scale redundancies, and the compact joint visual representation of low-level and medium-level features $\{c_L^{0-1}, c_R^{0-1}\}$ are generated, i.e.,

$$\{c_L^{0-1}, c_R^{0-1}\} = \text{SEU}(c_L^0 \oplus f_L^1, c_R^0 \oplus f_R^1) \quad (3)$$

where $\oplus$ denotes the channel-wise concatenation. Following that, the high-level stereo feature $\{f_L^2, f_R^2\}$ are merged with $\{c_L^{0-1}, c_R^{0-1}\}$ to obtain final compact joint visual representations $\{c_L^{0-2}, c_R^{0-2}\}$, which are then entropy encoded into bit-stream.

The decoder side of the SMFC module is symmetric with the encoder side. The reconstructed stereo multi-scale features $\{\hat{f}_L^i, \hat{f}_R^i | i \in 0, 1, 2\}$ are sequentially decoded from the reconstructed compact joint visual representation, i.e.,

$$\{c_L^{0-1}, c_R^{0-1}\}, \{\hat{f}_L^2, \hat{f}_R^2\} = \text{Split}(\text{SDU}(\hat{c}_L^{0-2}, \hat{c}_R^{0-2})) \quad (4)$$

$$\{\hat{c}_L^0, \hat{c}_R^0\}, \{\hat{f}_L^1, \hat{f}_R^1\} = \text{Split}(\text{SDU}(\hat{c}_L^{0-1}, \hat{c}_R^{0-1})) \quad (5)$$

$$\{\hat{f}_L^0, \hat{f}_R^0\} = \text{SDU}(\hat{c}_L^0, \hat{c}_R^0) \quad (6)$$

where $\text{Split}(\cdot)$ is the channel-wise split, and $\text{SDU}(\cdot)$ denotes the stereo decoder unit, which is composed of a bi-directional contextual transform module [11] and two transposed convolutions.

In contrast to separately dealing with stereo multi-scale features $\{f_L^i, f_R^i | i \in 0, 1, 2\}$, the SMFC progressively transforms the sparse stereo multi-scale features into the compact joint visual representation $\{c_L^{0-2}, c_R^{0-2}\}$, in which the spatial, inter-view as well as cross-scale redundancies are simultaneously reduced to improve the compression efficiency.

C. Loss Functions and Implementation Details

The proposed MVSFC-Net is trained in an end-to-end manner by the RD function, which comprises a task distortion term and a bitrate consumption term. To achieve optimal compression performance for the machine vision task, the task distortion term is calculated by the visual task loss, with the aim to reconstruct high-quality stereo multi-scale features for visual analysis. More specifically, we follow the method in [46] to calculate the visual task loss, which consists of three parts, including a focal loss for anchor classification $\mathcal{L}_{\text{cls}}$, a smooth L1 loss for box regression $\mathcal{L}_{\text{reg}}$, and a stereo focal loss on disparity $\mathcal{L}_{\text{dis}}$, i.e.,

$$D_v = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{dis}}. \quad (7)$$

Finally, the overall RD loss function can be summarized as follow:

$$\mathcal{L}_T = \lambda D_v + R_l + R_r \quad (8)$$

where the R_l and R_r denote the bitrate of the left image and the right image, respectively, and λ is the parameter to trade off the RD cost, which is set to $\{0.5, 1, 4, 16, 64, 256\}$. In terms of the detailed architecture of the MVSFC-Net, the stereo feature extraction module and visual analysis module are the same as those in [46]. More specifically, the backbone in the stereo feature extraction module is implemented by ResNet-34 [47]. The downsampling convolution in the proposed SEU is employed with a kernel size of 5×5 , a stride of 2, and a padding of 2, while the parameters of the transposed convolution in the SDU are the same as those of the downsampling convolution. The bi-directional contextual transform in the SEU and SDU is implemented following Lei et al. [11]. In the bi-directional contextual transform, two parallel paths containing a simplified non-local blocks [48], two convolution layers and the sigmoid function are utilized to generate the inter-view context, which is then exploited to transform the left and right features into more compact features. The visual analysis module consists of multi-stage fusion, box regression, and disparity estimation to obtain the final 3D object detection result.

IV. EXPERIMENTS

A. Experimental Configurations

1) *Datasets*: The MVSFC-Net is trained on the KITTI Object Detection Benchmark [49], where the stereo images are obtained by two color and two grayscale PointGrey Flea2 video cameras (10 Hz, resolution: 1392×512 pixels, opening: $90^\circ \times 35^\circ$). The KITTI Object Detection Benchmark has 7481 training and 7518 test images. Since the test ground-truth labels are not available, the KITTI training set is split into the training set (3712 images) and validation set (3769 images) following the Chen's split [50]. The top 100 pixels in the stereo images are cropped, and then the cropped stereo images are scaled to the resolution 288×1280 for fast inference and training.

2) *Training Strategy*: As the MVSFC-Net comprises several learnable components, it is hard to optimize the network stably by training from scratch. Therefore, a progressive training strategy is utilized to train the proposed MVSFC-Net. First, the stereo feature extraction module and visual analysis module are initialized by a YOLOStereo3D model [46], which is pretrained on the KITTI training set [49] for 80 epochs. Then, the MVSFC-Net is optimized for 20 epochs by the loss $\mathcal{L}_T$, with the parameters in the stereo feature extraction module and the visual analysis module fixed. Finally, all components in MVSFC-Net are end-to-end optimized together with the loss $\mathcal{L}_T$ for the final 80 epochs, which is helpful to improve the compression efficiency for visual tasks. In summary, the MVSFC-Net is optimized using Adam [51] with batch-size 4 for 180 epochs (i.e., $80 + 20 + 80$), where the learning rate is set as 1×10^{-4} , and decayed to 5×10^{-6} by the CosineAnnealingLR. The proposed MVSFC-Net is implemented with PyTorch and trained on a PC with Inter i9-12900K CPU @ 3.20 GHz and a GeForce RTX 3090 GPU.

B. Comparison Results and Analysis

1) *Comparison Algorithms and Evaluation Metrics*: To verify the efficiency of the proposed MVSFC-Net, ICM anchors recommended by MPEG (i.e., VVC image anchor [52] and VVC feature anchor [53]), ICM algorithm proposed in Wang's method [21], Liu's method [39], Kim's method [54], Andrade's method [55], and state-of-the-art learned SIC algorithms proposed in Lei's method [11] and Wodlinger's method [32] are utilized for comparison. As for the evaluation of the Wang's method [21], Lei's method [11], Wodlinger's method [32], and VVC image anchor [52], the whole stereo images are first compressed into bit-stream to calculate bitrate, and then the reconstructed image is input into the YOLOStereo3D model [46] to evaluate the performance of 3D detection task. For a fair comparison, the Wang's method [21], Liu's method [39], Lei's method [11], and Wodlinger's method [32] are retrained on the same KITTI training set as the proposed method. It is worthwhile to mention that both All-Intra configuration¹ and Low Delay P (LDP) configuration² under JVET common test conditions (CTCs) are tested for VVC image anchor [52], denoted as VVC-Intra and VVC-Inter, respectively. More specifically, the left image is compressed first and then the right image is compressed upon the compressed left image in the LDP configuration, while the left image and right image are compressed dependently in the All-Intra configuration. In this article, the performance of the VVC feature anchor [53], denoted as VVC-Feature, is evaluated by following the method suggested in [53]. In the VVC-Feature, the stereo multi-scale features are preprocessed, quantized, packed, and compressed by the VVC reference software VTM-12.0 [56], and the reconstructed features are sent to the visual evaluation to measure the 3D visual performance.

Regarding the evaluation metric, the widely adopted BD-rate [57] and BD-AP [16] are calculated to measure the coding performance. Specifically, the distortion is calculated by the accuracy of 3D visual task performance, including average precision for 3D box (AP_{3D}) and bird's eye view (AP_{BEV}). The bitrate is measured by the bits per pixels (BPP), which is calculated by the coding bits divided by the total number of pixels in the original images. In this article, the VVC-Intra [52] is set as the anchor to calculate the BD-rate and BD-AP.

2) *Coding Performance*: Table I presents the BD-rate as well as the BD-AP comparison results in terms of the AP_{3D} and AP_{BEV} . The positive value of the BD-rate indicates a rise in bit-rate consumption when compared to the anchor, while the negative value indicates a bit-rate saving. As shown in the table, the MVSFC-Net obtains an average of 81.005% BD-rate reduction against the anchor. In comparison, the Andrade's method [55], Kim's method [54], Lei's method [11], Wang's method [21], Liu's method [39], Wodlinger's method [32] and VVC-Inter [52] achieve 78.540%, 74.030%, 53.822%, 50.959%, 49.870%,

¹EncoderApp.exe -c encoder_intra_vtm.cfg -i input.yuv -b output.bin -o output.yuv -wdt 1248 -hgt 376 -q QP -InputChromaFormat=420

²EncoderApp.exe -c encoder_lowdelay_P_vtm.cfg -i input.yuv -b output.bin -o output.yuv -wdt 1248 -hgt 376 -q QP -InputChromaFormat=420 QP is set to 22, 27, 32, 37, 42, and 47.

TABLE I
COMPARISON RESULTS ON THE KITTI OBJECT DETECTION DATASET

Methods	Distortion Is Measured by 3D Detection AP						Distortion Is Measured by BEV Detection AP					
	Car		Pedestrian		AVG		Car		Pedestrian		AVG	
	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{BEV}	BD-Rate	BD-AP _{BEV}	BD-Rate	BD-AP _{BEV}
VVC-Feature [53]	418.728%	-24.685%	960.261%	-16.774%	689.495%	-20.730%	352.770%	-26.181%	890.801%	-18.384%	621.785%	-22.282%
VVC-Inter [52]	-28.462%	3.202%	-25.906%	1.344%	-27.184%	2.273%	-29.061%	3.699%	-26.190%	1.043%	-27.626%	2.371%
Wodlinger's method [32]	-58.686%	7.603%	-25.533%	1.858%	-42.110%	4.731%	-57.717%	8.405%	-19.218%	1.865%	-38.468%	5.135%
Liu's method [39]	-61.022%	6.971%	-38.717%	2.461%	-49.870%	4.716%	-60.302%	7.368%	-40.455%	2.378%	-50.378%	4.873%
Wang's method [21]	-52.876%	7.834%	-49.041%	2.704%	-50.959%	5.269%	-53.924%	9.347%	-55.558%	3.359%	-54.741%	6.353%
Lei's method [11]	-58.420%	7.796%	-49.225%	3.165%	-53.822%	5.480%	-62.560%	8.694%	-51.369%	3.503%	-56.965%	6.098%
Kim's method [54]	-72.214%	8.680%	-75.846%	4.340%	-74.030%	6.510%	-73.166%	9.904%	-75.873%	4.581%	-74.520%	7.243%
Andrade's method [55]	-76.089%	9.667%	-80.949%	3.982%	-78.540%	6.825%	-75.721%	10.917%	-75.592%	3.804%	-75.657%	7.361%
Proposed	-78.254%	12.477%	-83.756%	6.259%	-81.005%	9.368%	-76.347%	13.174%	-77.472%	6.709%	-76.910%	9.942%

Note: The VVC-Intra is set as anchor to calculate the BD-rate and BD-AP. Bold values indicates the best performance.

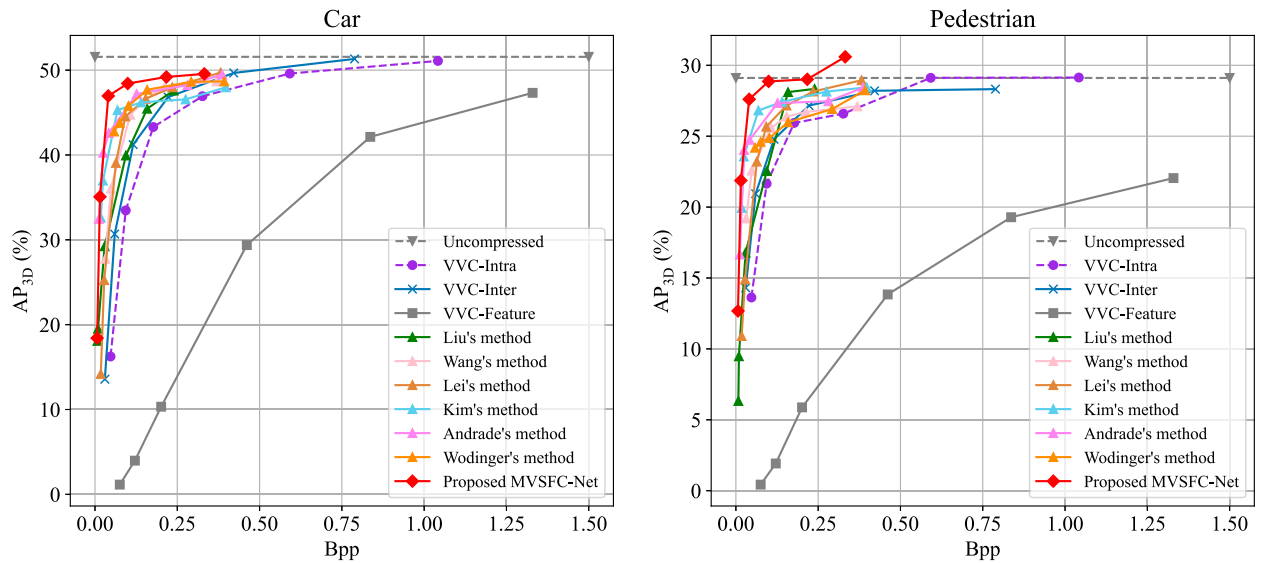


Fig. 2. RD curves comparison when the distortion is measured by AP_{3D} , where X axis denotes the BPP and Y axis denotes the average precision for 3D box (AP_{3D}). In the figure, the solid red line refers to the proposed MVSF-Net, whereas the blue dotted line and the other solid lines denote the anchor VVC-Intra and other comparative methods, respectively.

42.110%, and 27.184% BD-rate reduction, respectively, and the VVC-Feature [53] brings obvious BD-rate increase. Overall, the proposed MVSF-Net achieves higher performance than other comparison methods by a large margin, which strongly demonstrates the huge potential of SICM and the effectiveness of the proposed MVSF-Net. Consistent results are observed when the distortion is measured by the AP_{BEV} . As shown in the table, the proposed MVSF-Net also achieves the best compression efficiency among all comparison methods with an average of 76.910% BD-rate reduction as well as 9.942% AP_{BEV} improvement. Overall, the proposed MVSF-Net shows the best compression efficiency in SICM scenarios where machine task matters.

Figs. 2 and 3 show the RD curves comparison results when the distortion is measured by AP_{3D} and AP_{BEV} , respectively. It can be observed that the proposed MVSF-Net behaves better than all comparison methods on all bitrates. It is worth mentioning that the proposed MVSF-Net provides greater gains at lower bitrate, as the MVSF-Net only needs to compress feature-domain compact representation with lower information entropy.

3) *Visual Comparison:* To more intuitively prove the advantage of the proposed MVSF-Net, the visual comparison of 3D detection results in RGB image and 3D space are presented in Fig. 4. In the figure, the ground truth 3D bounding boxes are drawn with the blue bounding boxes, while the red bounding

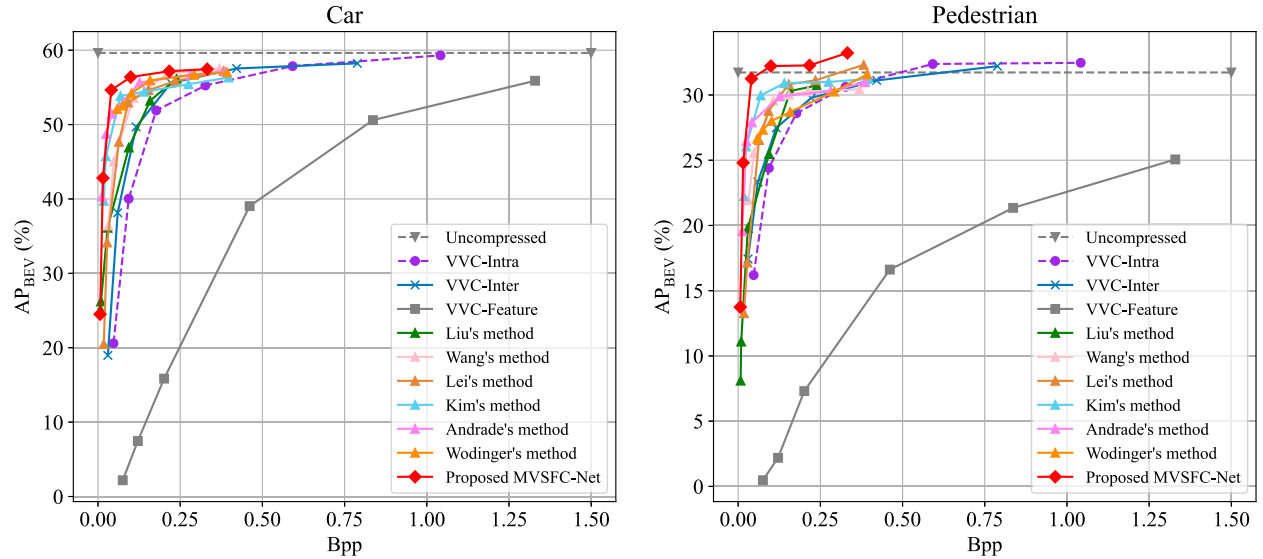


Fig. 3. RD curves comparison when the distortion is measured by AP_{BEV} , where X axis denotes the BPP and Y axis denotes the average precision for bird's eye view (AP_{BEV}). In the figure, the solid red line refers to the proposed MVSFC-Net, whereas the blue dotted line and the other solid lines denote the anchor VVC-Intra and other comparative methods, respectively.

boxes represent the 3D prediction results. As can be observed from the figure, compared with other comparison methods, the proposed MVSFC-Net is able to obtain more consistent 3D detection results with the uncompressed stereo images on both the Car category and the Pedestrian category, while having less bit-rate consumption.

C. Ablation Study

1) *Evaluation of SMFC*: The key of the proposed MVSFC-Net lies in efficient stereo features compression, as the stereo features exist enormously similar contents in images, view-points, and multiple scales. The SMFC module is designed to compress the stereo multi-scale features by effectively reducing the intra-view, inter-view as well as cross-scale redundancies. To verify the effectiveness of the SMFC module, the SMFC module is removed in the MVSFC-Net, denoted as “w/o SMFC.” In the “w/o SMFC,” the stereo multi-scale features are separately transformed into the latent representations by downsampling convolutions, and then the latent representations are concatenated together for entropy coding. As shown in Table II, when compared with the proposed MVSFC-Net, the network without the SMFC decreases the BD-rate reduction on average from 81.005% to 70.401%, and decreases the BD- AP_{3D} from 9.368% to 5.204%. Meanwhile, as shown in Fig. 5, at the higher bitrate, the network without the SMFC drops more detection accuracy and is even close to the RD curve of VVC-Intra [52]. This is because the inter-view and cross-scale redundancies in stereo multi-scale features are no longer effectively removed, resulting in a performance decrease.

2) *Evaluation of Multiple Redundancy Removal Operations*: To further validate the contributions of multiple redundancy removal operations in the proposed MVSFC-Net, the

inter-view redundancy removal (IRR) and CRR operations are progressively removed from the MVSFC-Net. Specifically, the IRR operation, including SEU, SDU and inter-view prior information in the entropy model, are first removed from the MVSFC-net, denoted as “w/o IRR.” On this basis, the CRR operation is removed from the model variant “w/o IRR,” which is defined as “w/o IRR&CRR.” In the “w/o IRR&CRR,” the stereo multi-scale features are separately transformed into the latent representations by downsampling convolutions, and then the latent representations are independently processed by entropy coding. As shown in Fig. 6 and Table III, when the IRR is removed from the MVSFC-Net, the BD-rate reduction decreases on average from 81.005% to 76.415%, and the BD- AP_{3D} decreases from 9.368% to 7.797%. This is mainly because that the inter-view redundancy in the stereo multi-scale features is no longer effectively removed. In addition, when the CRR is further removed from the model variant “w/o IRR,” the BD-rate reduction decreases on average from 76.415% to 63.884%, and the BD- AP_{3D} decreases from 7.797% to 5.462%, which is primarily attributed to the ignorance of cross-scale redundancy.

3) *Evaluation of Multi-Scale Feature Fusion Strategy*: To further evaluate the performance of the MVSFC-Net with different multi-scale fusion strategies, the concatenation operation is replaced by self-attention based feature fusion (SFF) proposed in [48], denoted as “MVSFC-Net w/ SFF.” As shown in Fig. 7 and Table IV, when compared with the proposed MVSFC-Net, the “MVSFC-Net w/ SFF” achieves nearly consistent rate-accuracy performance with higher computational complexity. To make a trade-off between coding performance and model complexity, the simple concatenation operations are chosen to facilitate the multi-scale feature fusion.

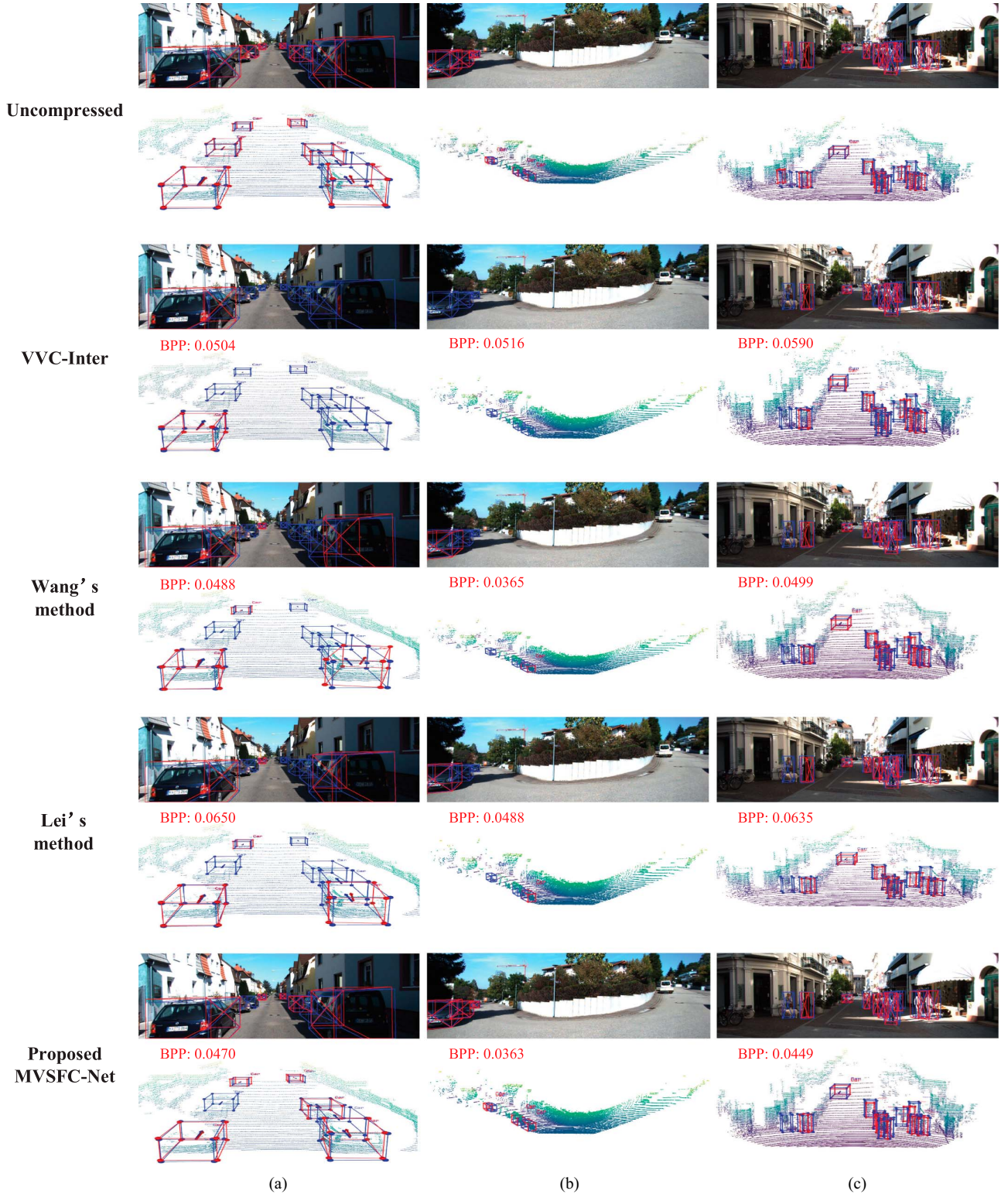


Fig. 4. Visual comparison of 3D detection results in RGB images and 3D space on the KITTI object detection dataset. The blue bounding box and red bounding box denote the ground truth result and predicted result, respectively. (a) 65th left image in the validation set. (b) 147th left image in the validation set. (c) 157th left image in the validation set.

4) *Evaluation of the Bi-Directional Contextual Transform Module:* To verify the effectiveness of bi-directional contextual transform module (Bi-CTM) [11] in SICM scenarios, the Bi-CTM is removed from the SEU and SDU in the proposed

MVSFC-Net, denoted as “w/o Bi-CTM.” In the “w/o Bi-CTM,” the right features are no longer fed to the left branch and the up non-local block as well as subsequent convolution layers are removed. That is, the left features are processed without

TABLE II
BD-RATE AND BD-AP_{3D} RESULTS OF MVSF-Net WITHOUT SMFC
MODULE ON THE KITTI OBJECT DETECTION DATASET

Model Variant	Car		Pedestrian		AVG	
	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}
w/o SMFC	-71.535%	6.473%	-69.266%	3.935%	-70.401%	5.204%
Proposed	-78.254%	12.477%	-83.756%	6.259%	-81.005%	9.368%

Note: Bold values indicates the best performance.

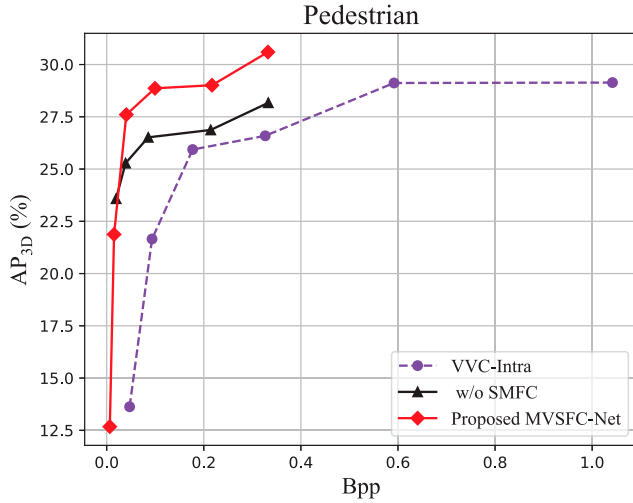


Fig. 5. RD curves comparison when the SMFC module is removed from the proposed MVSF-Net, where X axis denotes the BPP and Y axis denotes the average precision for 3D box (AP_{3D}). In the figure, the solid red line refers to the proposed MVSF-Net, whereas the blue dotted line and the black solid line denote the anchor VVC-Intra and model variant “w/o SMFC,” respectively.

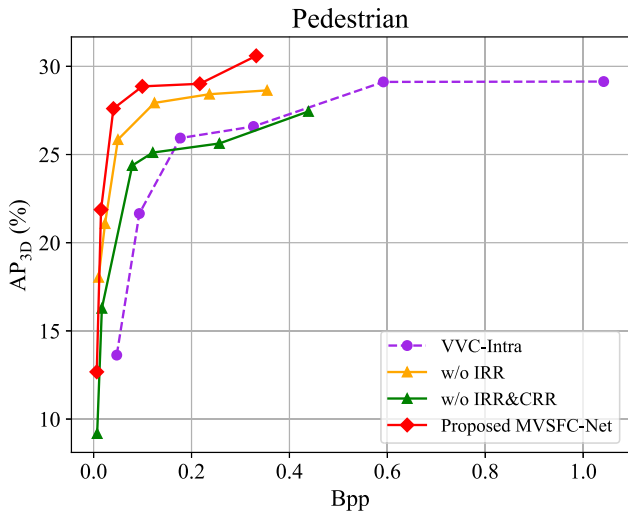


Fig. 6. RD curves comparison when the IRR and CRR operations are progressively removed from the proposed MVSF-Net, where X axis denotes the BPP and Y axis denotes the average precision for 3D box (AP_{3D}). In the figure, the solid red line refers to the proposed MVSF-Net, whereas the blue dotted line, yellow solid line and green solid line denote the anchor VVC-Intra, model variant “w/o IRR” and “w/o IRR&CRR,” respectively.

TABLE III
BD-RATE AND BD-AP_{3D} RESULTS OF MVSF-Net WITHOUT THE IRR
AND CRR OPERATIONS ON THE KITTI OBJECT DETECTION DATASET

Model Variant	Car		Pedestrian		AVG	
	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}
w/o IRR	-77.556%	10.791%	-75.273%	4.803%	-76.415%	7.797%
w/o IRR&CRR	-71.447%	9.136%	-56.320%	1.788%	-63.884%	5.462%
Proposed	-78.254%	12.477%	-83.756%	6.259%	-81.005%	9.368%

Note: Bold values indicates the best performance.

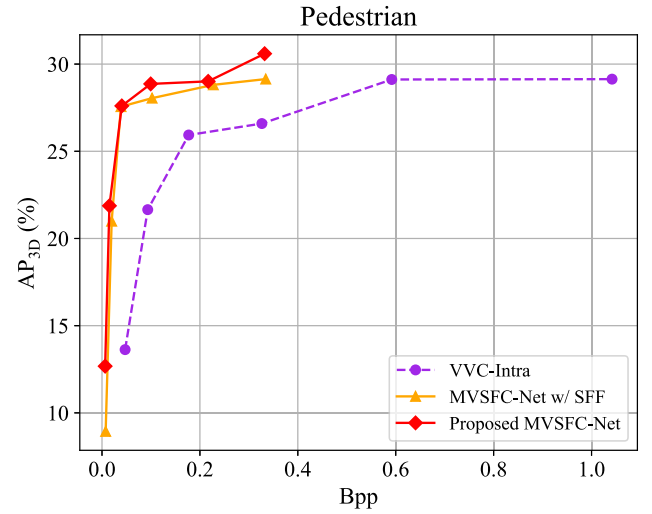


Fig. 7. RD curves comparison when the concatenation is replaced by SFF in the proposed MVSF-Net, where X axis denotes the BPP and Y axis denotes the average precision for 3D box (AP_{3D}). In the figure, the solid red line refers to the proposed MVSF-Net, whereas the blue dotted line and yellow solid line denote the anchor VVC-Intra and model variant “MVSF-Net w/SFF,” respectively.

TABLE IV
BD-RATE AND BD-AP_{3D} RESULTS OF MVSF-Net WITH THE SFF MODULE
ON THE KITTI OBJECT DETECTION DATASET

Model Variant	Car		Pedestrian		AVG	
	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}	BD-Rate	BD-AP _{3D}
MVSFC-Net w/ SFF	-79.989%	11.429%	-61.537%	5.780%	-70.763%	8.605%
Proposed	-78.254%	12.477%	-83.756%	6.259%	-81.005%	9.368%

Note: Bold values indicates the best performance.

concatenating any information about the right features. As shown in Fig. 8 and Table V, when the Bi-CTM is removed, the BD-AP_{3D} performance decreases from 9.368% to 8.329% and the BD-rate performance decreases from 81.005% to 70.576%. This is because the left features are not able to exploit the inter-view contextual information, resulting in an ineffective reduction of the inter-view redundancy.

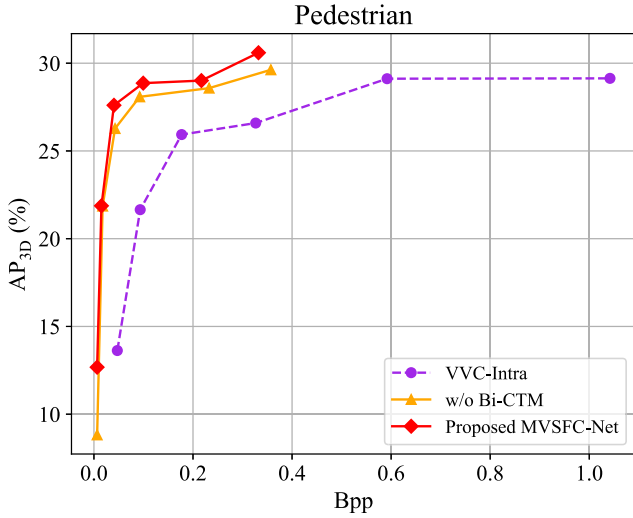


Fig. 8. RD curves comparison when the Bi-CTM is removed from the proposed MVSFC-Net, where X axis denotes the BPP and Y axis denotes the average precision for 3D box (AP_{3D}). In the figure, the solid red line refers to the proposed MVSFC-Net, whereas the blue dotted line and yellow solid line denote the anchor VVC-Intra and model variant “w/o Bi-CTM,” respectively.

TABLE V
BD-RATE AND BD- AP_{3D} RESULTS OF MVSFC-NET WITHOUT THE BI-CTM MODULE ON THE KITTI OBJECT DETECTION DATASET

Model Variant	Car		Pedestrian		AVG	
	BD-Rate	BD- AP_{3D}	BD-Rate	BD- AP_{3D}	BD-Rate	BD- AP_{3D}
w/o Bi-CTM	-65.819%	11.386%	-75.333%	5.272%	-70.576%	8.329%
Proposed	-78.254%	12.477%	-83.756%	6.259%	-81.005%	9.368%

Note: Bold values indicates the best performance.

D. Computational Complexity Analysis

We compare the computational complexity of the proposed MVSFC-Net with comparison algorithms in terms of number of floating-point operations (FLOPs), parameters, and encoding/decoding time. As shown in Table VI, compared with the Lei’s method [11] and Wang’s method [21], although the proposed MVSFC-Net has more parameters, the FLOPs of the proposed MVSFC-Net is reduced obviously, which is owing to the dealing with the feature-domain compact representation rather than the pixel-domain stereo images. As for the measurement of the encoding and decoding time, the QP in the traditional methods (i.e., VVC-Feature [53], VVC-Intra [52], and VVC-Inter [52]) is set to 37, and the practical arithmetic coding is adopted in learning-based methods (i.e., the proposed MVSFC-Net, Lei’s method [11], and Wang’s method [21]) for a fair comparison. As shown in Table IV, the proposed MVSFC-Net only consumes 0.396 s to encode the stereo images, and requires the lowest encoding time among all comparison methods. As for the decoding time, the proposed method requires lowest decoding time when compared with learning-based methods, and has a higher decoding time than traditional methods due to the auto-regressive model in the entropy coding.

TABLE VI
COMPARISON RESULTS OF COMPUTATIONAL COMPLEXITY

Method	FLOPs (G)	Params. (M)	Enc-Time (s)	Dec-Time (s)
VVC-Feature [53]	—	—	365.716	0.080
VVC-Intra [52]	—	—	46.324	0.051
VVC-Inter [52]	—	—	37.480	0.048
Liu’s method [39]	288.913	172.916	1.213	2.930
Wang’s method [21]	515.934	124.188	0.601	4.864
Lei’s method [11]	1172.725	136.722	1.283	5.384
Proposed	418.533	181.171	0.396	1.666

TABLE VII
COMPARISON RESULTS ON THE MIDDLEBURY DATASET

Method	BD-Rate (%)	BD-PSNR (dB)
Liu’s method [39]	-61.358	0.093
Kim’s method [54]	-86.097	0.67
Proposed	-97.065	1.996

Note: The Lei’s method is set as anchor to calculate the BD-rate and BD-PSNR. Bold values indicates the best performance.

E. Discussion

1) *Evaluation of the Generalization Ability of MVSFC-Net:* To further evaluate the generalization ability of the proposed MVSFC-Net in different 3D visual tasks and scenes, the experiments have been conducted on the indoor scene dataset Middlebury [58], [59], [60] to verify the effectiveness of the MVSFC-Net in the stereo low-light enhancement task. Referring to [61], 136 image pairs are collected from the Middlebury dataset for training and 36 image pairs for testing. Besides, the dual-view low-light enhancement network (DVNet) proposed in [61] is utilized to initialize the stereo feature extraction module and visual analysis module. To verify the efficiency of the proposed MVSFC-Net on the stereo low-light enhancement task, two ICM algorithms proposed in Liu’s method [39], Kim’s method [54] and learned SIC algorithm proposed in Lei’s method [11] are chosen for comparison, where Lei’s method [11] is set as the anchor to calculate the BD-rate and BD-PSNR [57].

Table VII shows the BD-rate as well as the BD-PSNR comparison results on the Middlebury dataset. As shown in the table, the MVSFC-Net obtains 97.065% BD-rate reduction against the anchor. In comparison, the Liu’s method [39] and Kim’s method [54] achieve 61.358% and 86.097% BD-rate reduction, respectively. The experimental results in Table VII show that the proposed MVSFC-Net achieves the highest RD performance in the stereo low-light enhancement task, which proves the effectiveness of the proposed MVSFC-Net. In addition, RD curves in terms of the PSNR metric are shown in Fig. 9. It can be observed that the proposed MVSFC-Net yields better performance than other comparison methods on all bitrates. This is mainly because that the proposed MVSFC-Net is able

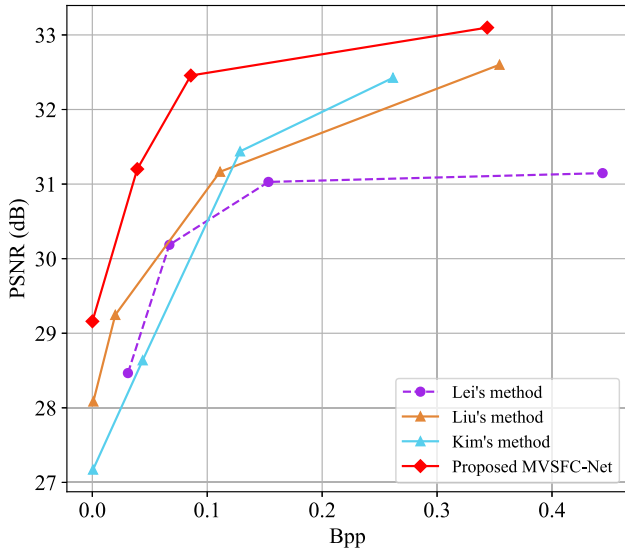


Fig. 9. RD curves comparison when the distortion is measured by PSNR on the Middlebury dataset, where X axis denotes the BPP and Y axis denotes PSNR. In the figure, the solid red line refers to the proposed MVSFC-Net, whereas the blue dotted line and the other solid lines denote the anchor Lei's method and other comparative methods, respectively.

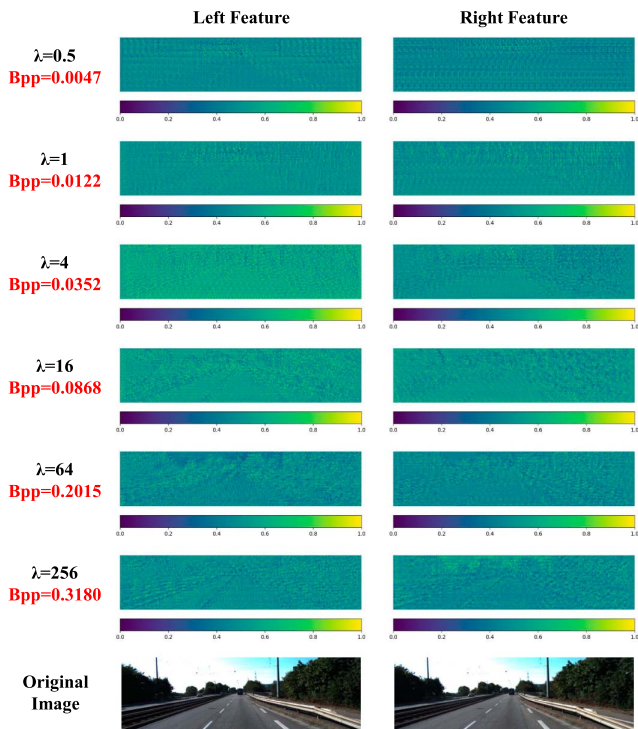


Fig. 10. Visualization results of the large-scale left and right features coded as bit-stream at different bit-rates. The feature maps presented above are all extracted from the first image in the validation set.

to jointly reduce spatial, cross-scale and inter-view redundancies. In addition, benefiting from the proposed SMFC module, the proposed MVSFC-Net is capable of transforming stereo

multi-scale features into a joint visual representation by modeling intra-view, cross-scale and inter-view correlations, which effectively enhances the robustness and adaptability of the MVSFC-Net.

2) *Evaluation of the Disparity Compensation Ability of SEU*: To further validate the disparity compensation ability of the proposed SEU, we also provide visualization results of the large-scale left and right features encoded as bit-streams at different bit-rates. Specifically, give a specific feature map, an averaging operation is performed along the channel dimension to obtain a 2D feature map, which is then normalized into the interval $[0, 1]$ to acquire the final visualization result. The feature visualization results are shown in Fig. 10. It can be observed from the visualization results that at high bit-rates, the proposed SMFC module is able to achieve disparity compensation, which is mainly because the proposed method effectively promotes inter-view interaction via the bi-directional conditional context transform module. At low bit-rates, the high compression level leads to a large number of similar flat regions in the left and right features, making it difficult to identify the inter-view disparity compensation.

V. CONCLUSION

This article first explores the potential framework for SICM. In particular, a MVSFC-Net is proposed to extract, compress, and transmit compact visual features in a task-driven manner. In the MVSFC-Net, a SMFC module is proposed to remove spatial, inter-view, and cross-scale redundancies simultaneously by progressively generating the compact visual joint representation. Experimental results prove that the MVSFC-Net presents more advanced compression capability in terms of the visual task performance, when compared with both state-of-the-art learning-based method and the latest traditional video coding standard VVC. In the future, the proposed MVSFC-Net can be further extended by developing transform codecs or entropy models specific to 3D visual task attributes.

DATA AVAILABILITY STATEMENT

The code is available at <https://github.com/zhaob10/MVSFC-Net-for-SICM>.

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